

Dynamics of a Parametrically Excited Pendulum with Finite-Range Impacts

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Introduction

Parametric resonance occurs when periodic modulation of a system parameter produces large oscillations. It plays a critical role both in preventing instability in engineered structures and in high-precision technologies such as quadrupole ion traps and mass spectrometers used in proteomics and atmospheric sensing.

The canonical model is the vertically driven pendulum described by the Mathieu equation, which predicts regions of stability and exponential growth. However, real systems often involve impacts or constraints, where this description may fail.

We study this effect in the clackers toy: two balls connected by a string with a vertically driven midpoint. Between impacts the motion follows a parametrically driven pendulum, while collisions introduce non-smooth dynamics, providing a minimal system for testing the limits of Mathieu theory under realistic conditions.



Mathieu equation

The motion of a vertically driven pendulum is described in dimensionless form by

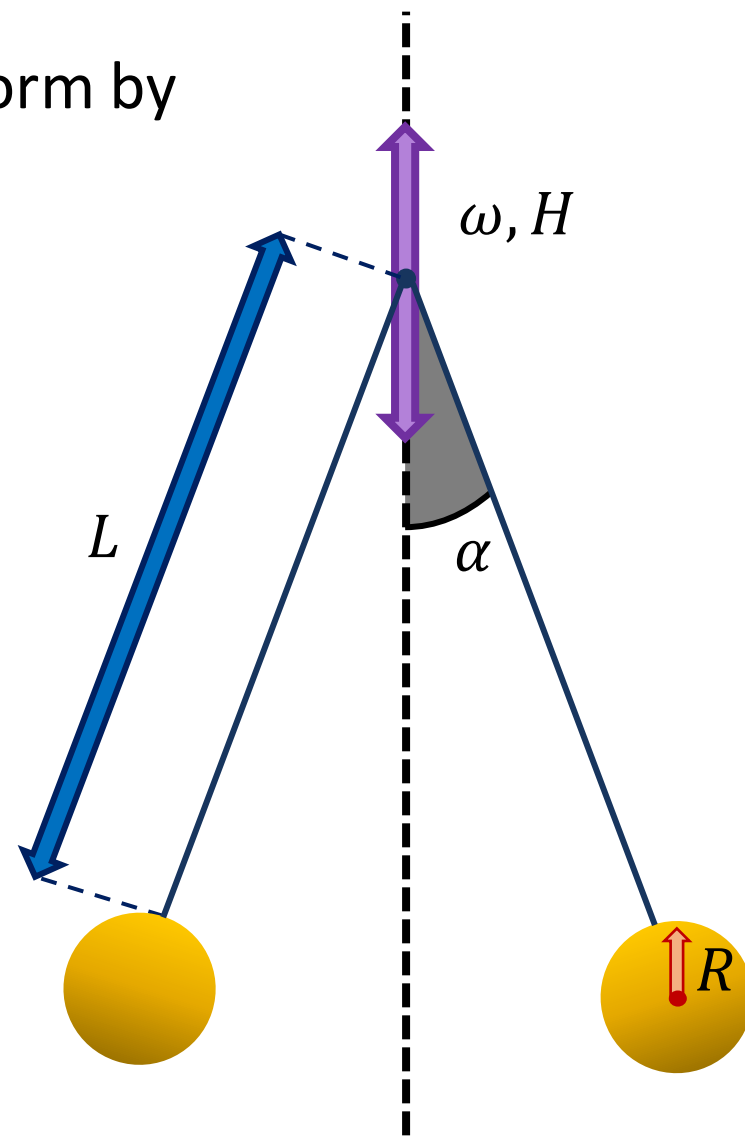
$$\ddot{\alpha}(\tau) + 2\beta\dot{\alpha}(\tau) + \left(\frac{1}{\delta^2} + \epsilon \sin \tau\right) \sin \alpha(\tau) = 0.$$

Parameters

$$\delta = \omega/\omega_0 \quad \epsilon = H/L$$

where δ is the dimensionless driving frequency and ϵ is the dimensionless driving amplitude.

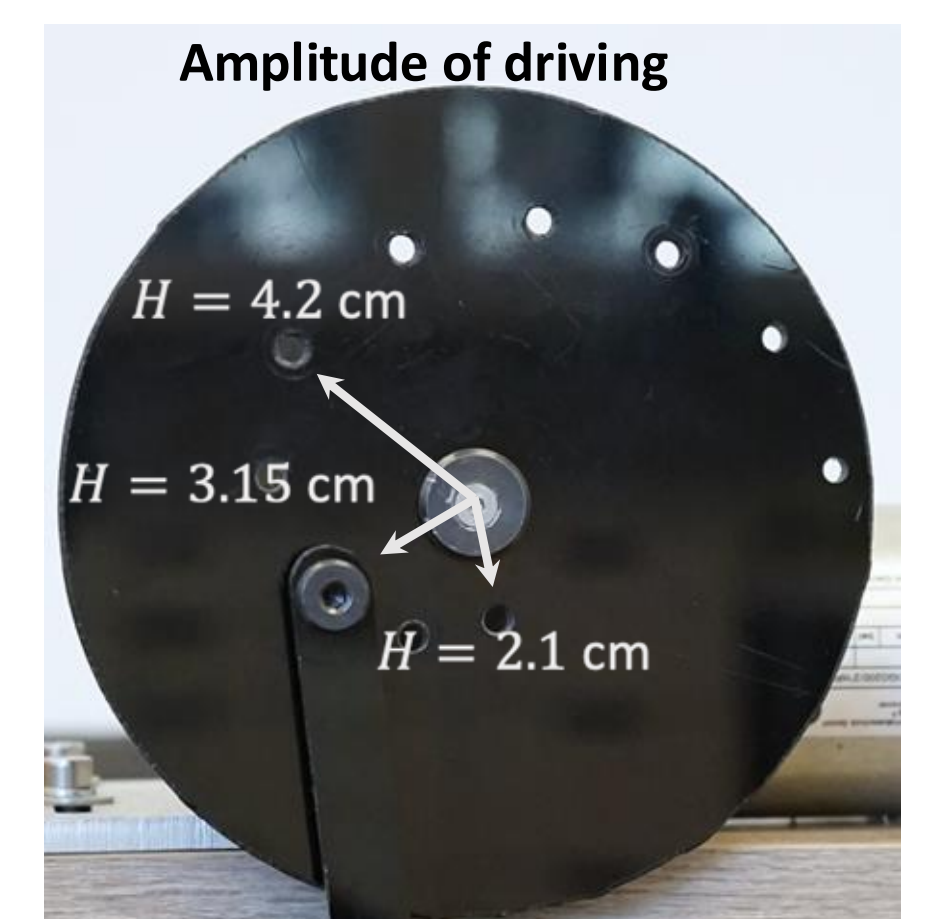
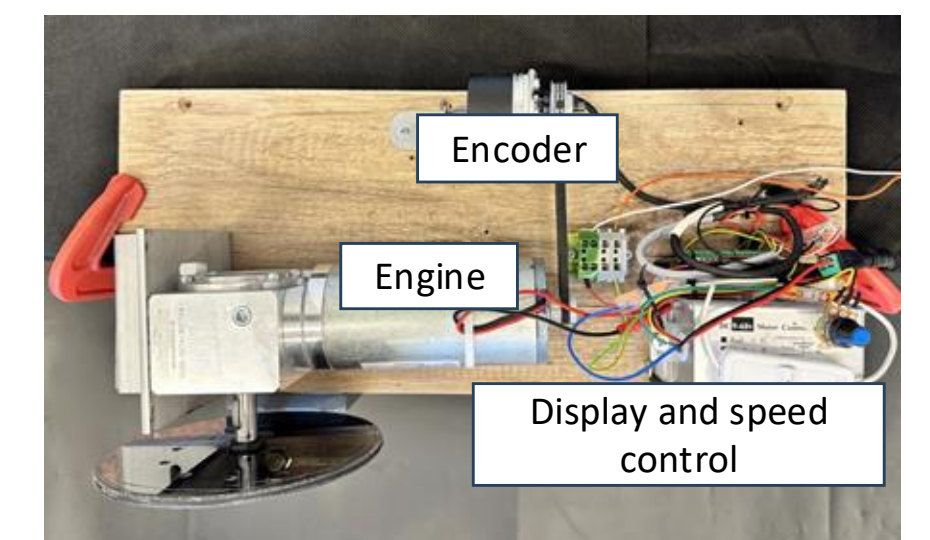
For small oscillation angles, this equation reduces to Mathieu equation. Its solutions include a stable non-oscillatory regime and parametric resonance, with the response determined mainly by δ and ϵ . The boundaries between these regimes are typically obtained from stability analysis based on Floquet theory.



Experimental setup

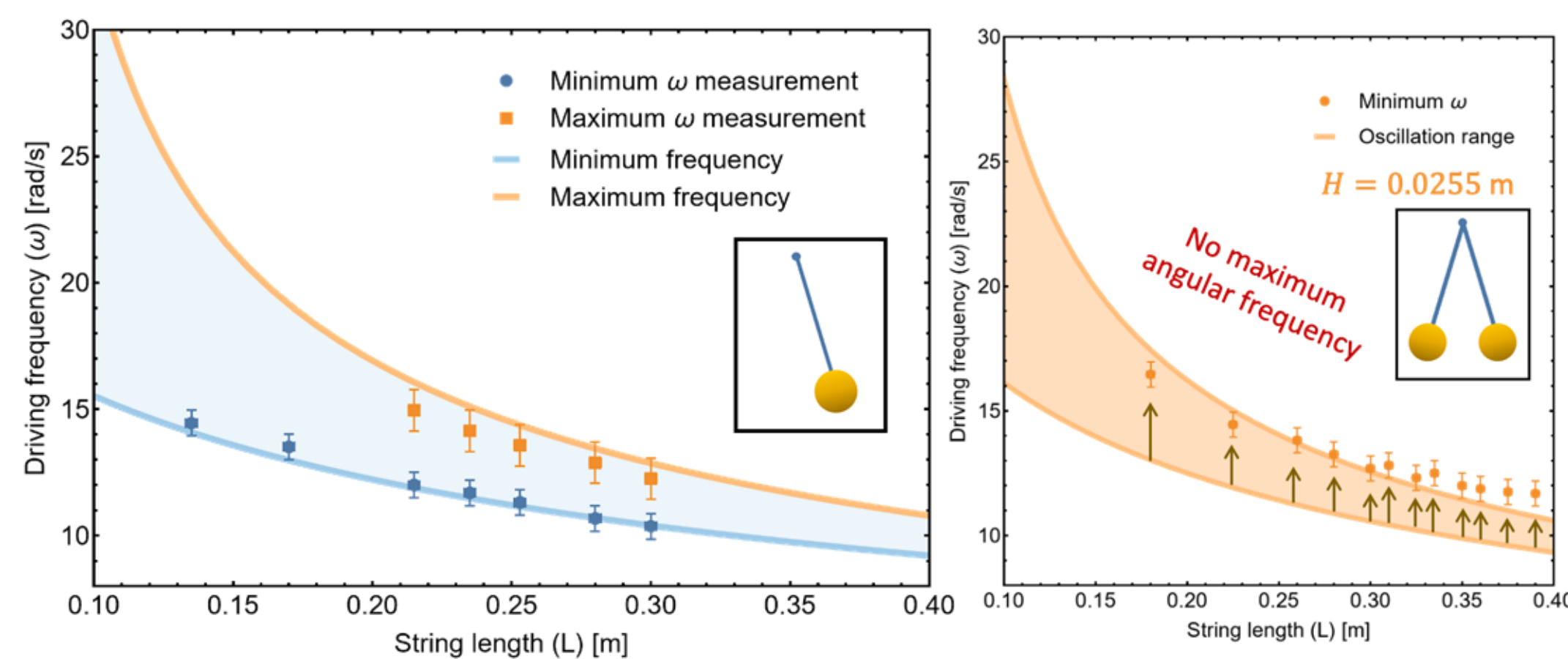
Experimental setup allowed real time modifications of driving frequency thanks to responsive engine-encoder system. It could produce driving frequencies in range (0 – 25) Hz, with driving amplitudes between 2.1 and 4.2 cm set by an eccentric mechanism. Additionally, changing clackers used, we modified coefficient of restitution (COR), string length L and ball radius R .

We filmed clackers oscillations after a long equilibration period following initialization of motion and automatically tracked string positions to extract angular position as a function of time. Like the numerically collected data, experimental amplitude measurements were obtained using the RMS method applied to the experimental trajectories.



Limits of the Mathieu model

The first plot compares the stability boundaries predicted by the Mathieu equation with the measured resonance range of a single parametrically driven pendulum, defined by the lowest and highest driving frequencies at which resonance occurs.



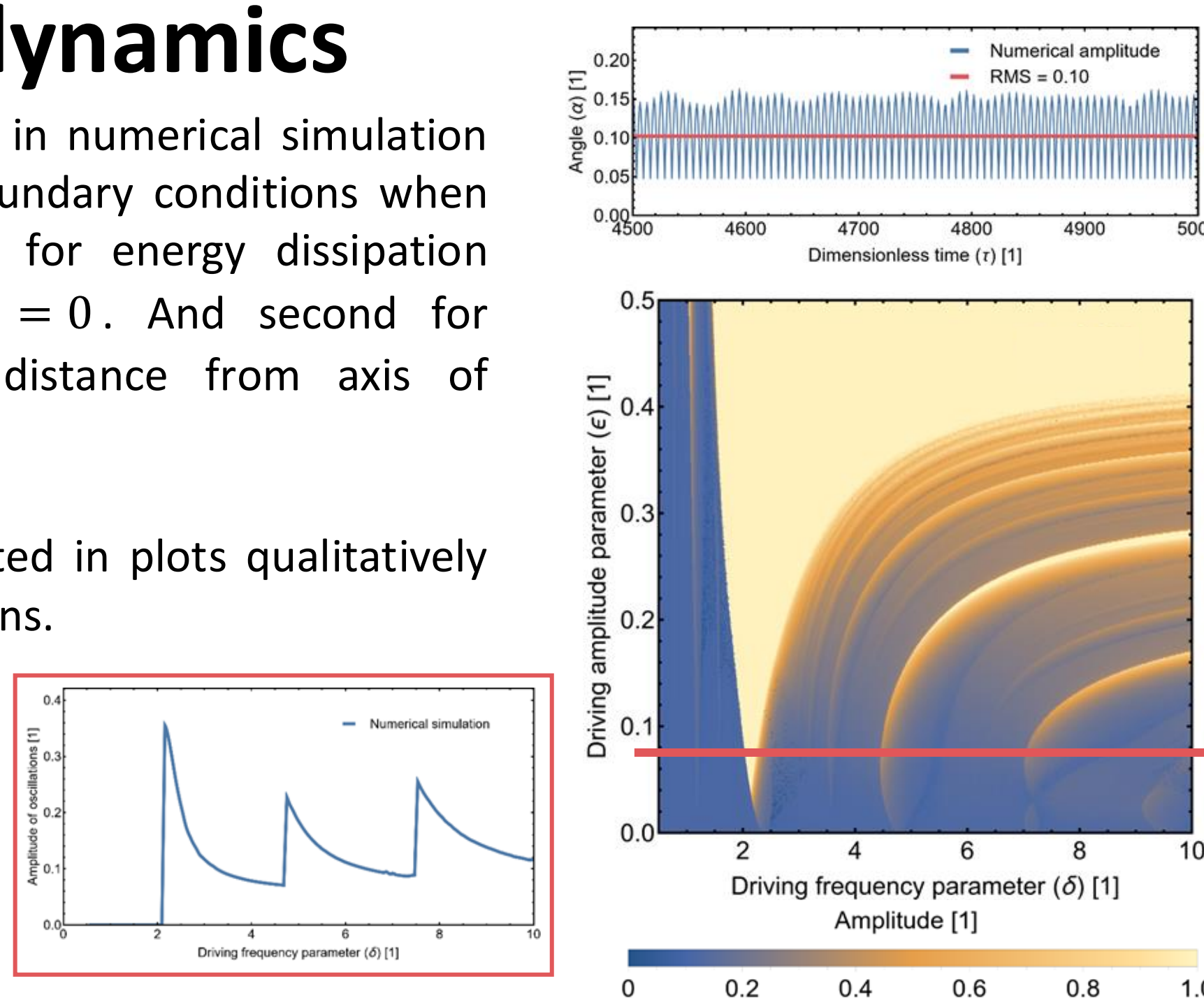
The agreement is reasonable for the single pendulum. In the clackers system, however, collisions shift the onset of oscillations to higher frequencies and eliminate the upper frequency limit. This highlights the limitations of the Mathieu model and motivates extending it to include collisions.

Collision-induced dynamics

We investigated influence of collisions in numerical simulation by separately introducing different boundary conditions when modelling collisions. One responsible for energy dissipation losses ($\dot{\alpha}(t) \rightarrow -e \dot{\alpha}(t)$) when $\alpha(t) = 0$. And second for collisions happening at $\alpha(t) = \alpha_c$ distance from axis of symmetry, then ($\dot{\alpha}(t) \rightarrow -\dot{\alpha}(t)$).

Only second type of simulations resulted in plots qualitatively differing from Mathieu equation solutions.

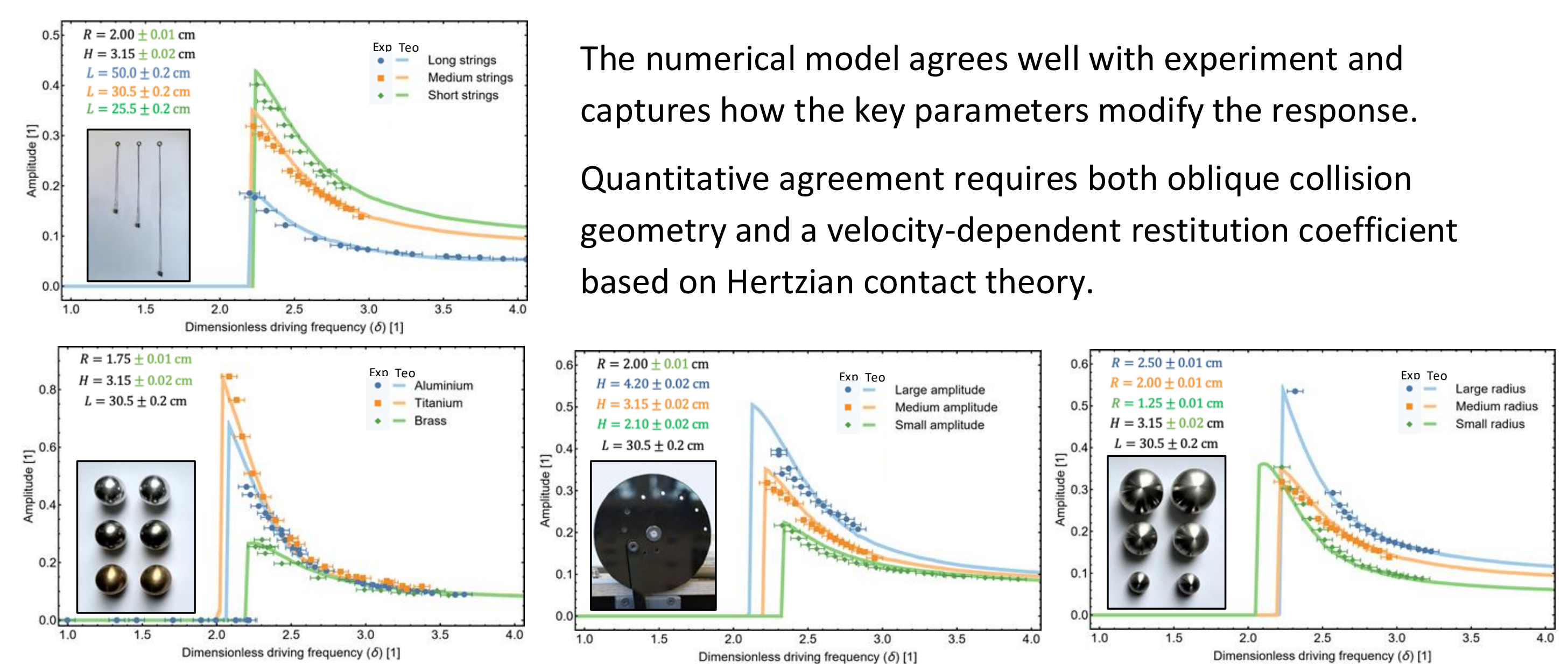
For those we could observe first resonance peak to be shifted and non-zero amplitude solutions for all frequencies above first resonant frequency.



Quantitative agreement with experiment

The numerical model agrees well with experiment and captures how the key parameters modify the response.

Quantitative agreement requires both oblique collision geometry and a velocity-dependent restitution coefficient based on Hertzian contact theory.



Key Findings

- The Mathieu equation provides a useful baseline for the parametrically driven pendulum, but it does not fully describe the clackers system once collisions are included.
- Finite-size effects cause collisions to occur away from the axis of symmetry, giving rise to a third parametric response: stable finite-amplitude oscillations not predicted by the Mathieu model.
- Dissipation mainly introduces second-order corrections, reducing the oscillation amplitude and shifting the resonance peaks.
- Including both oblique collision geometry and a velocity-dependent restitution coefficient is necessary for quantitative agreement with experiment.
- The extended model reproduces the measured response over a broad range of driving frequencies and system parameters, including sphere radius, string length, driving amplitude, and material-dependent restitution.

References

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